

Scalability & Future Plan

8.Project Demonstration | Credit Card Approval Prediction

Prepared For	Prepared By	Domain	Date
Credit Card Approval Prediction	Ankitha Vana	Finance	July 1, 2026

Scalability Plan

- Deploy model to IBM Watson Machine Learning for scalable API-based scoring.
- Containerize the Flask application for repeatable deployment.
- Add batch upload support for screening many applicants at once.
- Store prediction logs in a database for audit and monitoring.
- Automate model retraining when new labeled application data is available.

Future Enhancements

Enhancement	Value
Explainable AI	Show feature contributions behind each prediction.
Fairness checks	Monitor whether predictions are biased across demographic groups.
Admin dashboard	Track approvals, rejections, model metrics, and application trends.
Database integration	Persist users, applicants, credit histories, and predictions.
API security	Add authentication and role-based access for bank staff.
CI/CD pipeline	Automate testing and deployment through GitHub Actions.